How the Soft-Lock Finds a Signal in the Noise

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The soft-lock finds where NAVTEX characters start by testing 14 possible alignments against the signal's built-in repetition, and it commits only when one alignment wins clearly over about six seconds of evidence.

The problem

NAVTEX (SITOR-B) sends 100 bits a second as 7-bit characters. Two properties matter. Every character is sent twice, the repeat arriving 35 bits (350 ms) after the first copy. And every valid character contains exactly four 1s and three 0s.

Nothing in the stream marks where a character begins. The receiver must find the bit position (7 choices) and which of two interleaved groups holds first copies and which holds repeats (2 choices). That gives 14 hypotheses. In weak reception many individual bits are wrong, so deciding hard 1/0 values first and aligning afterwards fails. The soft-lock postpones that decision.

Scoring each alignment

The demodulator passes on a soft value for every bit: the sign is the likely bit and the size is the confidence. The soft-lock keeps up to a minute of them.

Every 140 ms it makes one comparison per hypothesis. It takes a 7-value group and the group 35 bits later, and asks whether they look like two copies of one character. Two measures are added, each scaled against what pure noise produces:

- **Agreement:** the cosine of the angle between the two groups. It is close to +1 when the copies agree and about 0 for unrelated data.
- **Validity:** the groups are added bit by bit, so noise partly cancels while the signal reinforces. The sum is matched against the 32 valid codes and the best match is taken.

A wrong alignment compares unrelated stretches of signal and scores near zero on average. The right one scores a little above zero each time, so the evidence has to be accumulated.

Detecting the lock

Scores are added over the last 45 comparisons (6.3 s) and expressed as a z-score: how many standard deviations above what noise alone would give. The lock is declared when the best hypothesis reaches z = 5 and leads the runner-up by at least 1.

The threshold is a trade-off. In 24 hours of recorded noise, z = 5 produced no false locks, z = 4 about 0.3 an hour and z = 3.5 about 1.8 an hour.

Detection takes several seconds, but no text is lost. The soft values from just before the window are still stored, and decoding replays them under the new alignment.

Decoding the characters

The two copies of each character are combined before any decision is made. Their soft values are added bit by bit and one decision follows, so a bit lost to noise in one copy is often rescued by the other.

- Four 1s in the result: accepted.
- Three or five 1s: one bit from valid, so the least certain bit is flipped.
- Anything else: the copies are judged separately. One valid copy is used. Two valid but different copies give `!`, and no valid copy gives `~`.

Characters are held back for about 2 s before output, so that later evidence can still withdraw them.

Staying locked

All 14 hypotheses keep being scored after the lock, which allows two things.

- **Release.** If the locked alignment and its two one-bit neighbours all stay below z = 1.5 for 5 comparisons in a row (0.7 s), the signal is judged to have gone. The lock is released, the held-back characters are discarded as stray tail, and the line ends, so the next message starts on its own line.
- **Following.** If a different alignment is clearly ahead for 3 checks in a row, the lock moves to it. A one-bit slip, where the bit clock has drifted, needs $z \geqslant 4$ and a lead of 2. An unrelated alignment, such as a second, stronger station, needs $z \geqslant 5$ and a lead of 3. To find where the change happened, the decoder tries every split point between the old and new scores, picks the best fit, and decodes again from there.

Filtering junk, and results

A phasing (idle) signal read out of step can produce plausible characters. Two filters remove them. A run of eight or more identical letters is dropped. Three "idle look-alike" pairs in a row are dropped too: the two copies of a real character agree, so real text cannot look like this. The 2 s delay lets the start of such a run be withdrawn once it is recognised.

In the project's synthetic tests, 10% of characters were recovered at about −13 dB signal-to-noise with the soft-lock, against about −8 dB for the standard decoder. Residual garbage near weak signals and where stations overlap is accepted.

Note: dB figures are signal-to-noise ratios, with the noise measured in a 2,500 Hz bandwidth.